

A Big Data Framework for Price Tracking and Product Matching in Sri Lankan E-Commerce

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Chapter 3

Problem Specification and Requirements

Currently, hunting for deals and comparing prices across multiple online stores can be a time-consuming and tedious task for consumers. With the increasing number of retailers and e-commerce marketplaces offering their products to be purchased online, consumers often find themselves overwhelmed with the sheer volume of options available to them. Moreover, it is difficult for consumers to identify whether sales and discounts offered on these platforms are genuine or if they are simply marketing tactics to attract customers, where the original prices are marked up to create the illusion of a discount. With the same product offered at different platforms, consumers are often left wondering which platform offers the best deal, and whether they are getting the best value for their money.

While LLMs and agentic systems have to a certain extent, have proven useful in solving these problems, they do have their limitations.

- **Limited selection:** When prompted, LLMs do not provide results from all available platforms, but rather to a selection of 3-5 platforms, which may not be representative of the entire market. This can lead to consumers missing out on better deals available on other platforms.
- **Inability to track historical prices:** While they are able to access real-time data, LLMs are unable to track historical prices of products, which can be useful in determining whether a sale or discount is genuine or not.
- **High cost and resource usage:** LLMs are resource-intensive and can be costly to run, especially when dealing with large volumes of data. While for small-scale product comparison, LLMs may be feasible, for larger-scale comparisons, they may not be practical.

3.1 Aims and objectives

3.1.1 Aims

This project aims to address these limitations by introducing a framework that can be used to identify products and track their prices across multiple online platforms, while also providing historical price data to help consumers make informed decisions about their purchases. It will also include a product-matching algorithm that can identify the same product across different platforms, even if they are listed under different

names or descriptions. The framework will be designed to be scalable and efficient, allowing for large-scale price tracking and cross-site product comparisons without incurring high costs or resource usage.

This framework will be generalizable, allowing it to be used across different online platforms, without the definition of rules or logic specific to any one platform.

While this framework is intended to be re-usable and adaptable to different product categories and platforms, this project will implement a proof-of-concept implementation, which will be a web-based application focused on Sri Lanka's e-commerce platforms, specifically focusing on consumer electronics.

3.1.2 Objectives

Primary objectives

- To design a distributed web crawler than can efficiently crawl multiple e-commerce platforms and collect page data.
- To evaluate different web-page classification techniques and use the most effective one to identify product pages from other types of pages.
- To evaluate different web-page information extraction techniques and use the most effective one to extract relevant product information from product pages.
- To implement a price-tracking framework to track historical prices for a given product at a specific retailer.
- To evaluate different product-matching techniques that can identify the same product across different platforms, and use the most effective one to implement a cross-site product comparison system.

Secondary objectives

- To implement a boilerplate identification and removal algorithm to minimize the amount of irrelevant data needs to be stored and processed.
- To implement an observability framework to monitor the performance of the system and identify any issues that may arise.
- To implement a web-based application that allows users to search for products and view their historical prices across different platforms, as well as compare prices and identify the best deals available.

3.2 Requirements

3.2.1 Functional requirements

- Provided a set of seed URLs, the system must crawl each URL, render the page, and store its HTML content in a BLOB storage.

- For each page crawled, the system must identify hyperlinks to other pages and add them to the crawling queue.
- The system must be able to classify each crawled webpage as a product page or a non-product page.
- For each product page, the system must extract the product's title, current price, product image, and whether the product is in stock or out of stock.
- The system must be able to display the price-history of a product over time.
- Given a specific product from a specific e-commerce website, the system must be able to find the same product offered on other e-commerce websites.

3.2.2 Non-functional requirements

- The system must be able to track at least 100,000 products from at least 30 different e-commerce websites.
- The price of each product must be updated at least once every 72 hours.
- The distributed crawler must ensure that the crawling process does not impact the performance of the target e-commerce websites, and must respect the robots.txt file of each website.
- The system must be able to retrieve the price history for a product in under 500 milliseconds.
- The system must be able to find the same product offered on other e-commerce websites in under 2 seconds.
- The system must be implemented at a low cost of under USD 300 per month, including cloud hosting and storage costs.

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